Functional Data Analysis

James O. Ramsay, McGill University, Montreal, QC, Canada
Bernard W. Silverman, University of Bristol, Bristol, UK

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Abstract

Most statistical analyses involve one or more observations taken on each of a number of individuals in a sample, with the aim of making inferences about the general population from which the sample is drawn. In an increasing number of fields, these observations are curves or images. Curves and images are examples of functions, since an observed intensity is available at each point on a line segment, a portion of a plane, or a volume. For this reason, we call observed curves and images 'functional data,' and statistical methods for analyzing such data are described by the term 'functional data analysis.' It is the smoothness of the processes generating functional data that differentiates this type of data from more classical multivariate observations. This smoothness means that we can work with the information in the derivatives of functions or images. This article includes several illustrative examples.

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The goals of FDA are essentially the same as for other branches of statistics, and include the following:

1. to represent and transform the data in ways that aid further analysis,
2. to display the data so as to highlight various characteristics,
3. to study important sources of pattern and variation among the data, and
4. to explain variation in an outcome or dependent variable by using input or independent variable information.

We shall illustrate the nature of functional data, these goals, and FDA tools available or under development, through a series of examples.

Human Growth Data: Looking at Velocity and Acceleration

Figure 1 shows the growth patterns of 10 Swiss boys involved in the Zurich Longitudinal Growth Study (Falkner, 1960). Each boy's height is measured at 29 separate ages, but it is more instructive to think of this as a sample of 10 growth curves or functions than as a set of 290 numerical observations. Of particular interest in the study of growth is the way that growth speeds up and slows down at various stages of development, and this is much better illustrated by studying the velocity and acceleration of height. Figure 2 gives estimates of the acceleration or second derivative of these 10 height functions, and shows that around puberty there is a strong pulse of acceleration in growth, followed by sharp deceleration. This is the pubertal growth spurt (PGS), and there is also often a smaller spurt, hardly visible in Figure 1 but obvious in Figure 2, at around age 6.

The Mean Function and the Registration Problem

How would one estimate an overall acceleration curve to quantify the size of these spurts? The dashed curve in Figure 2 is the average acceleration over the sample at each time point, but the PGS for this average is smaller in size and longer in duration

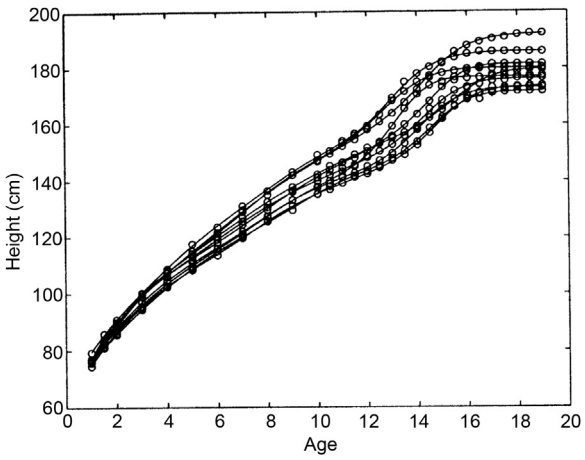


Figure 1 The heights of 10 Swiss boys measured at 29 ages. Notes: Points indicate unequally spaced ages of measurement; solid line is a function fitted to these data by monotone smoothing.

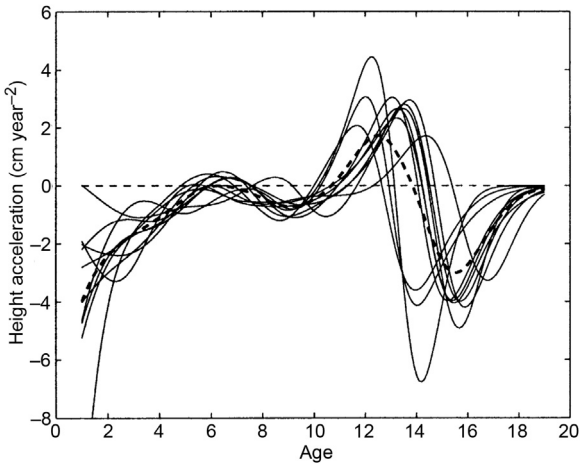


Figure 2 The estimated accelerations of height for 10 Swiss boys. Notes: Heavy dashed line is cross-sectional mean. This is a rather poor summary of the curves because of the presence of the timing of the pubertal growth spurt, called phase variation.

than that of any of the individual spurts. Averages are supposed to look like typical observations; so what went wrong?

The problem is that the curves vary in two ways, termed amplitude variation and phase variation. Alternatively, these are called vertical and horizontal variation, respectively, since amplitude concerns the height of a function at a given point, and phase the location in time of a feature such as an acceleration peak. In short, the actual acceleration peaks in the PGS vary in both size and timing. An artificial example demonstrating these two sorts of variation is presented in **Figure 3**. The phase shift displayed in this figure is constant across the timescale, but many data sets require more complicated and nonlinear transformations of the time axis in order to align curve features like the peaks and valleys in **Figure 3**. The presence of phase variation arises in FDA because many processes involve a system time, or in this case biological time, that does not unfold at the same rate as clock time. In this case, the biological clock of each boy marks off time at its own unique rate.

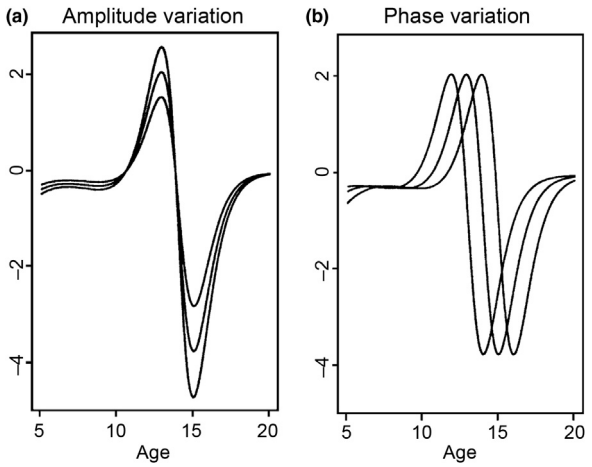


Figure 3 Panel (a): three height acceleration curves varying only in amplitude; Panel (b): three curves varying only in phase.

A key part of the FDA methodology is time-warping or registration of the data curves to align specific features or to minimize variability, since estimating the mean and most other analyses are thrown off by the presence of phase variation. The time-warping functions may be simple constants, linear functions, or more complicated, nonlinear functions. The amount or shape of warp may be of great interest (see Ramsay and Silverman, 1997: Chapters 5 and 8). **Figure 4** shows time-warping functions and registered height acceleration curves for the growth curves of **Figure 1** estimated using a method developed by Ramsay and Li (1998). The time-warping curve for each individual shows his maturation pattern. A warping function consistently above the diagonal dashed line implies that clock time lags behind development, and therefore development is late in clock terms. If height is measured relative to biological age, the curves in **Figure 2** are aligned as shown in **Figure 5**. Differences in height acceleration after registration correspond to pure amplitude variation in growth

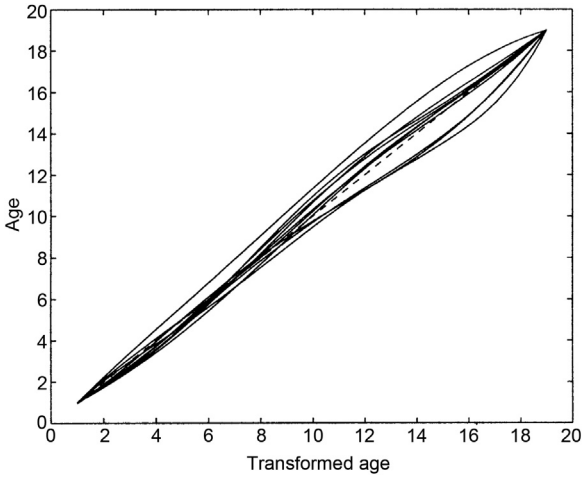


Figure 4 Time-warping functions taking chronological age on the horizontal axis into biological age on the vertical axis.

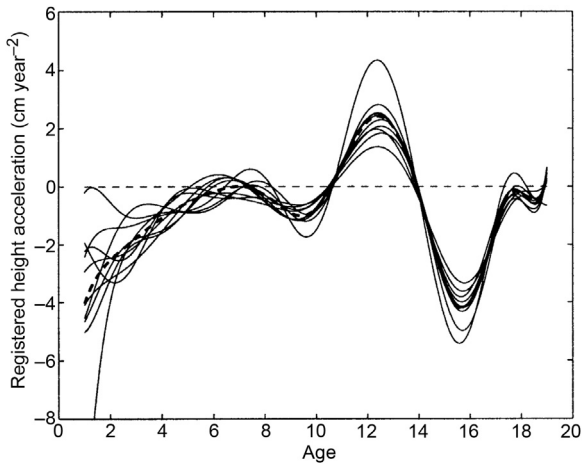


Figure 5 The height acceleration curves in **Figure 2** aligned or registered by measuring height with respect to the biological time shown in **Figure 4**. Notes: Heavy dashed line is the mean acceleration. This provides a much better summary of the actual curves than **Figure 2**.

patterns. The mean of the registered curves in Figure 5 is a much better summary than the mean of the unregistered data in Figure 1.

The Nondurable Goods Index and More Derivatives

Valuable information may be gained by plotting one derivative of a functional observation against another, a technique called phase/plane plotting. The US index of nondurable goods production is an important economic indicator, and considerable insight may be gained by plotting the acceleration of this index against its rate of increase. If the index had a smooth trend plus a sinusoidal seasonal variation, this plot would be approximately circular with suitable choice of scales. The typical year 1970 in Figure 6 has three main cycles, of which the spring cycle is rather smaller than the summer and fall/winter cycles. Comparisons of plots for different years is instructive. During the stable period 1964–66, there is little year-to-year change, but the period 1973–75 in Figure 6 shows considerable instability, due primarily to the end of the Vietnam War. A typical plot for a more recent year, 1994, shows that the whole pattern has changed, indicating structural changes in the methods and economics of production. For further details, see Ramsay and Ramsey (2002).

Functional Principal Components Analysis

Many FDA methods are adaptations of classical multivariate methods such as principal components analysis (PCA), linear modeling, and analysis of variance (see also Linear Hypothesis; Multivariate Analysis: Discrete Variables, Overview). Functional PCA demonstrates the way in which a set of functional data varies from its mean, and, in terms of these modes of variability, quantifies the discrepancy from the mean of each individual functional datum. Figure 7 (Figure 6.2 in Ramsay and Silverman, 1997) shows the first four principal component weight curves for temperature records for 35 Canadian weather stations. The first mode of variability is overall temperature, but with larger effects in the winter; the second corresponds to variation in annual range of temperature; the third partly to a time-shift of the overall cycle, and the fourth to short, hot summers and long, mild winters, or vice versa. Individual weather stations can then be characterized according to their scores on these four criteria. The interpretability of the components can often be improved by a suitable rotation of the original principal components; see Ramsay and Silverman (1997: Chapter 6).

A Functional Linear Model and Regularization

An example of functional linear modeling is provided by the study of the dependence of the acceleration of the lower lip (lipacc) in speech on neural activity, as measured by electromyographical (EMG) recording. It is reasonable to suppose that the neural activity at time s continues to affect lip movement for only a short period of time, which we denote by δ and this leads to a model of the form

$$\text{lipacc}(t) = \alpha(t) + \int_{t-\delta}^{t} \text{EMG}(s)\beta(s,t) + \varepsilon(t)$$

developed by Malfait et al., (2000). The bivariate function $\beta(s, t)$ plays the same role as the regression coefficient in standard linear regression, and quantifies the relationship between the

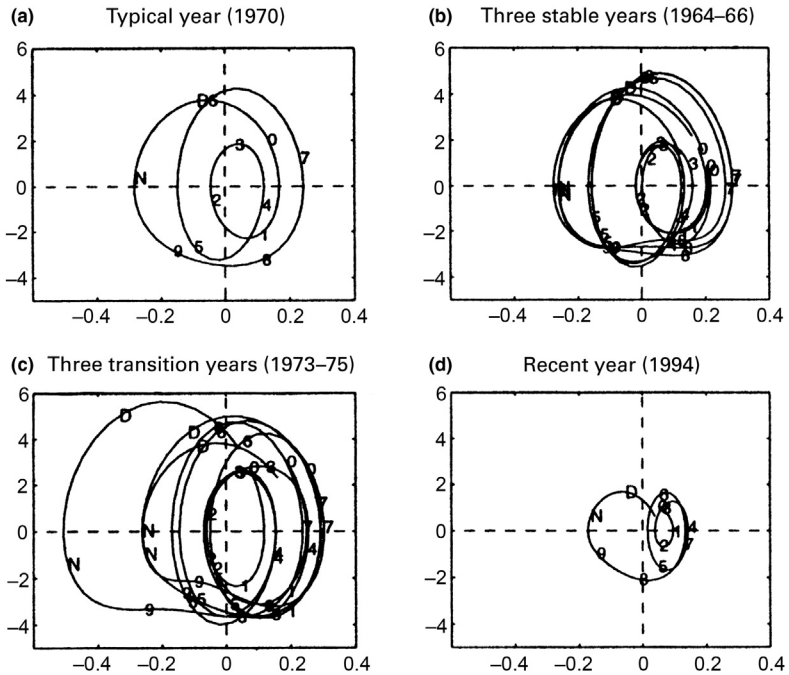


Figure 6 A phase/plane plot of *US index of nondurable goods production*. Each panel plots second derivative on vertical axis against first derivative on horizontal axis. Plots show structure of seasonal variation in index, and how structure varies with time.

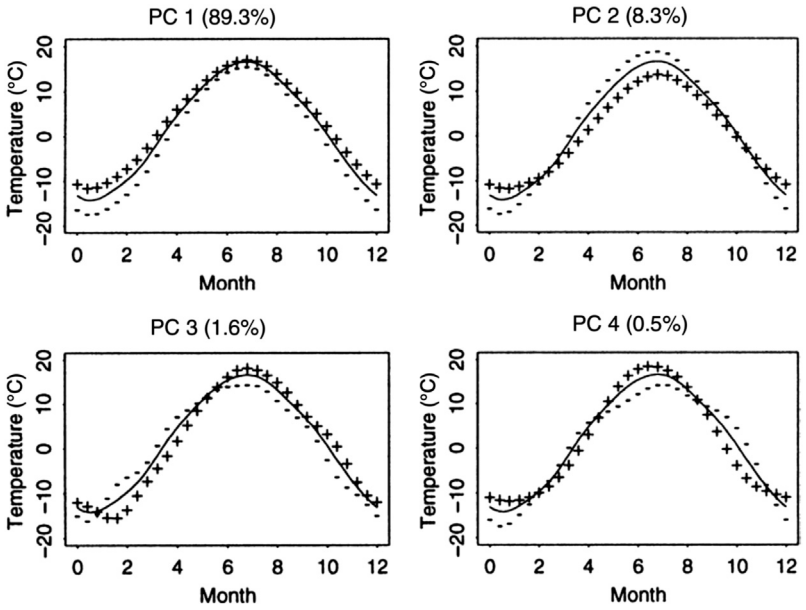


Figure 7 A principal components analysis of mean temperature curves for 35 Canadian weather stations. Notes: Solid line is the mean curve taken across all stations. The effects of adding (+) and subtracting (−) a suitable multiple of each component are also shown.

independent functional variable EMG, and the dependent functional variable lip. One particular issue is the appropriate choice of the time lag δ the length of time within the model that neural activity can continue to affect movement. Models for increasing values of δ correspond to nested linear models, and in this case a study of the amount of variation explained demonstrates that there is little effect beyond a lag of about half a second.

An important theoretical and practical issue is the necessity or otherwise of incorporating some form of smoothing or regularization into a functional analog of a classical multivariate procedure. In principal components analysis, the principal components, such as those displayed in **Figure 7**, are smoothed or regularized and in some procedures, such as canonical correlation analysis (see also Multivariate Analysis: Discrete Variables, Overview) or certain types of linear modeling for the lip/EMG data, regularization is necessary to give any meaningful results at all; see Leurgans et al., (1993), and Ramsay and Silverman (1997: Chapters 11 and 12).

Modeling with Derivatives: A Central Theme

It is the smoothness of the processes generating functional data that differentiates this type of data from more classical multivariate observations. This smoothness means that we can work with the information in the derivatives of functions or images. We looked at acceleration in the growth data, found something interesting in the joint variation of the velocity and acceleration of nondurable goods production, and used derivatives to define roughness penalties; see Ramsay and Silverman (1997: Chapter 4) for the weather data.

We may also use derivatives to construct models for data. Such models are called differential equations, and are part of the standard model-building toolkit in the natural sciences and engineering. Differential equation models are often called dynamic models, since they are especially effective at representing processes unfolding in time. Ramsay (2000) developed a linear differential equation for handwriting involving velocity, acceleration, and the third derivative of the tip of the pen. **Figure 8** shows a sequence of pen positions during the writing of the word 'statistics' in Mandarin, and also shows as a solid line its approximation on the basis of the dynamic equation estimated from 50 such samples of Chinese script. The equation not only captured the essential features of this particular sample, but was also able to model a fair amount of the sample-to-sample variation as well. And, finally, it also provided an excellent account of the velocity, acceleration, and jerk functions at the same time. Ramsay and

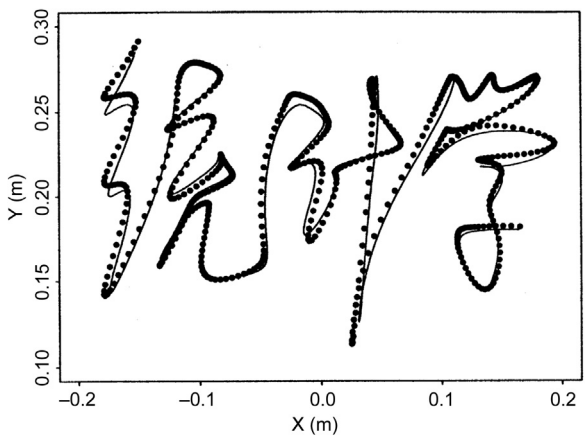


Figure 8 Observations of the tip of the pen while writing the word 'statistics' in Mandarin. Notes: Dots indicate 2400 observations; solid line is an approximation based on a second-order linear differential equation.

Silverman (1997: Chapters 13 and 14) take up this topic of dynamic modeling with functional data, including an account of a technique called 'principal differential analysis,' developed by Ramsay (1995).

Relationships with Other Branches of Statistics

Of course, statisticians have been exploring curves for a very long time, and there are many older statistical methodologies that can continue to inform and enrich an FDA. First of all, FDA usually begins with some kind of nonparametric regression (e.g., see also Exploratory Data Analysis: Univariate Methods) in order to replace discrete and possibly noisy curve values by smooth, continuous functions. Density estimation also involves data smoothing. Typically, time series analysis (Brillinger, 1981; Box and Jenkins, 1994) considers long sequences of curve values corresponding to equally spaced time values that are more or less stationary; that is, the data show little overall trend, and the covariance among neighboring values does not change much from one time region to another. However, methods for nonstationary processes are also under active development, and there are strong structural links between times series methods such as state-space modeling and FDA methods such as smoothing. Longitudinal data analysis or multilevel modeling (Diggle et al., 1994; Searle et al., 1992) usually looks at shorter sequences of curve values sampled over many cases, and studies the structure of between- and within-curve variation. It seems likely that this field and FDA will merge in a number of useful ways. Multivariate statistics has many methods developed specifically for functional data that have preceded FDA, but which have not explicitly exploited the smoothness of functional data or made other uses of derivatives (see also Probability Density Estimation; Longitudinal Data: Event–History Analysis in Discrete Time; Multivariate Analysis: Discrete Variables, Overview).

Conclusion

By the smoothness of a functional observation we have meant the possibility that one or more derivatives, such as velocity or acceleration, may be examined or used in the modeling process. Otherwise, FDA makes only minimal assumptions about the structure of variation within and between curves. We have seen, however, that teasing apart amplitude and phase variation is a central issue, and one that has largely been ignored until recently. Certain types of models considered to be fairly routine and well understood in multivariate statistics, such as linear regression, have many more variants in the functional domain, and are being researched actively. Software for FDA is also being developed, and it is likely that FDA will soon be considered as a routine item in a data analyst's black bag of instruments and nostrums. Further information on FDA and related issues may be obtained in Pinheiro and Bates (2000), Ramsay and Silverman (1997), and Simonoff (1996).

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